

Design and Implementation of an IMU Sensor Network for Real Time Human Kinematic Reconstruction in Yoga Practice for Elderly

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Abstract—Unguided home exercise, particularly among the aging population, presents a significant risk of musculoskeletal injury due to a lack of professional oversight and proprioceptive awareness. Conversely, clinical environments lack accessible and objective tools for quantifying dynamic joint angles, traditionally relying on expensive and easily obstructed optical camera systems. This paper presents an affordable wearable sensor network engineered to bridge the gap between independent home safety and clinical diagnostics. Utilizing a multiplexed three sensor Inertial Measurement Unit (IMU) architecture placed at the chest, biceps, and forearms, the system captures high speed kinematics. Sensor telemetry is processed using 4D quaternion mathematics to prevent Gimbal Lock and is mapped in real time to a Unity based digital twin. The software enforces strict orthopedic constraints and employs a finite state machine (FSM) to deliver visual biofeedback against an expert baseline avatar. Furthermore, the system functions as a standalone clinical tool featuring native OS integration to extract high fidelity timestamped kinematic datasets in CSV and JSON formats. Experimental results demonstrate stable 115200 baud telemetry, robust hardware watchdog recovery, and successful artifact rejection, validating the system's readiness for both preventative home use and downstream predictive AI diagnostics.

Index Terms—Inertial Measurement Unit (IMU), BNO055, quaternion kinematics, digital twin, wearable sensor network, yoga rehabilitation, elderly biomechanics, Unity3D, finite state machine, range of motion.

NOMENCLATURE

IMU	Inertial Measurement Unit
FSM	Finite State Machine
ROM	Range of Motion
SLERP	Spherical Linear Interpolation
GUI	Graphical User Interface
ESP32	Espressif Systems 32-bit Microcontroller
I ² C	Inter-Integrated Circuit (serial protocol)
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
RMSE	Root Mean Square Error

I. INTRODUCTION

A. The Home Practice Hazard

The integration of physical therapies, such as yoga, into daily routines is highly beneficial for the aging population; however, social anxiety and accessibility barriers frequently limit seniors' ability to engage in professionally guided sessions [1]. As a result, patients often resort to unguided home practice. Lacking professional oversight and adequate proprioceptive awareness, users routinely breach safe biomechanical limits. Unsupervised practice has been associated with a threefold increase in the rate of severe trunk and spinal injuries among older demographics [2], [3]. Without immediate corrective feedback, minor postural deviations can compound into severe musculoskeletal trauma.

B. The Diagnostic Gap in Clinical Settings

While home practice suffers from a lack of guidance, clinical practice suffers from a lack of accessible, objective telemetry. Physical therapists and orthopedists require reliable tools to quantify joint angles and Range of Motion (ROM) during dynamic movements. Historically, high fidelity kinematic tracking has been restricted to optical motion capture laboratories [4]. These systems are prohibitively expensive, require dedicated laboratory space, and are highly susceptible to visual occlusion when limbs cross. Consequently, routine in-clinic assessments remain highly subjective, relying primarily on visual estimation or manual goniometry, which limits the ability to track long term incremental rehabilitation progress.

C. Proposed Approach

To address these dual challenges, this paper proposes a private, low cost wearable tracking system designed for dual use in personal and clinical environments. By engineering a custom IMU network that continuously tracks upper body kinematics, the system bypasses the limitations of optical occlusion. For the patient, it delivers a real time, constraint driven 3D avatar that provides immediate proprioceptive visual feedback to enforce safety limits. For the clinician, it serves

as a robust diagnostic pipeline, generating automated clinical report cards and exporting raw timestamped data for further analysis.

The remainder of this paper is organized as follows. Section II details the hardware and software architecture. Section III describes the clinical assessment pipeline. Section IV presents experimental results, and Section V concludes with directions for future work.

II. PROPOSED SYSTEM ARCHITECTURE

A. Hardware Architecture and Sensor Design

The hardware framework consists of a customized three sensor IMU network utilizing BNO055 absolute orientation sensors (Bosch Sensortec), each operating with a 16-bit accelerometer, a 16-bit gyroscope, and an integrated magnetometer. To ensure reliable anchoring and zero slip operation against key anatomical landmarks, specifically the chest, biceps, and forearms, custom 3D printable ergonomic mounts were engineered (see Fig. 1 and Fig. 2). By relying on gravity vectors and quaternion mathematics rather than line of sight cameras, the telemetry system is rendered entirely immune to visual occlusion while strictly preserving patient privacy.

To facilitate the simultaneous tracking of multiple limb segments without I²C bus address conflicts (the BNO055 supports only two fixed I²C addresses), a TCA9548A 8-channel multiplexer was integrated into the sensor network. Sensor fusion and data processing are executed at the edge utilizing an ESP32 microcontroller (240 MHz dual-core Xtensa LX6), which streams formatted telemetry to the host machine via a high speed serial pipeline at 115200 baud.

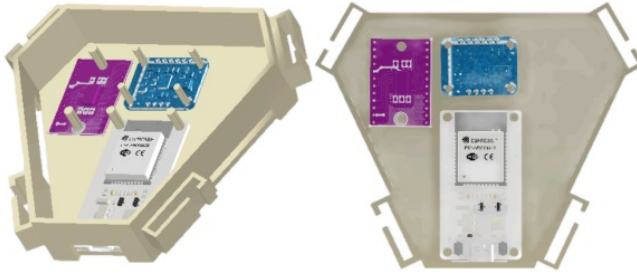


Fig 1 : Design for chest piece.

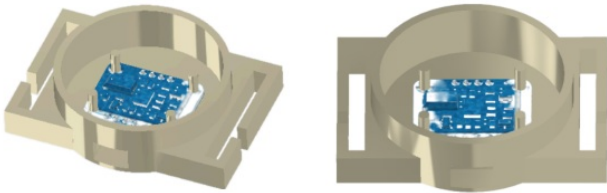


Fig 2 : Design for forearm and bicep piece

Fig. 1. CAD render of the custom 3D printable ergonomic sensor mounts designed for zero slip anchoring at anatomical landmarks (chest, biceps, forearms).

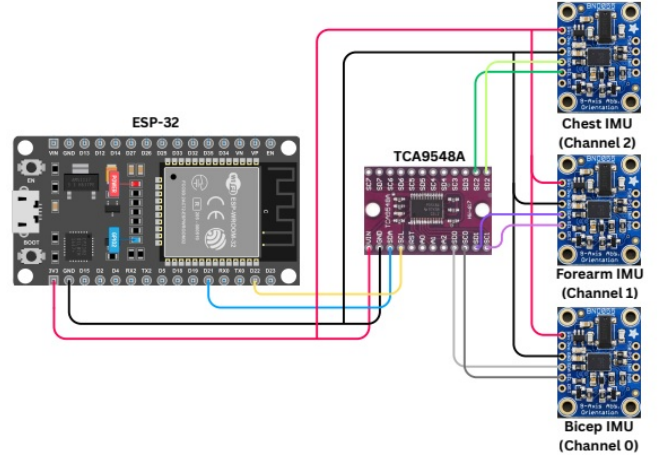


Fig. 2. Photograph of the assembled sensor network worn by a subject, illustrating correct sensor placement at the chest, biceps, and forearm anatomical landmarks.

B. Kinematic Engine and Software Integration

The central software environment operates within the Unity 6 engine. Raw sensor telemetry is processed using 4D quaternion mathematics [5] to map rotational data onto a rigged 3D humanoid mesh. Representing orientation as a unit quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]^T$ with $\|\mathbf{q}\| = 1$ actively prevents Gimbal Lock and graphical distortion during complex movements such as overhead shoulder flexion.

Strict biomechanical safety limits were programmed directly into the engine, acting as a *digital shock absorber*. The system mathematically rejects anatomically impossible joint angles, thereby filtering erroneous sensor noise and enforcing clinical orthopedic tolerances [6], [7].

To mitigate inherent IMU sensor noise and transient magnetic interference [8], the kinematic engine employs Spherical Linear Interpolation (SLERP) for all simulated joint rotations [9]. SLERP between two unit quaternions $\mathbf{q}_1$ and $\mathbf{q}_2$ is defined as:

$$\text{SLERP}(\mathbf{q}_1, \mathbf{q}_2; t) = \mathbf{q}_1 (\mathbf{q}_1^{-1} \mathbf{q}_2)^t, \quad t \in [0, 1] \quad (1)$$

Additionally, an algorithmic transient filter continuously monitors consecutive frame angular velocity. It autonomously rejects anomalous data spikes that exceed physiological human speed limits, ensuring a medically accurate digital twin.

To align the physical and virtual coordinate frames, a programmable dynamic calibration protocol known as the *Tare* system was implemented. This requires the user to assume a standardized neutral pose, overriding default magnetic alignments to lock the patient's real world orientation to the digital space.

C. Application Deployment and Portability

The Unity 6 application was compiled into a standalone Windows executable (.exe) packaged as a self contained compressed archive (ZIP). It requires no proprietary drivers, external dependencies, or complex installation procedures.

A clinician or patient can simply extract the ZIP file onto any standard Windows PC and immediately run the application. Adaptive deterministic UI scaling ensures the interface remains clinically accessible across hardware ranging from standard 1080p laptops to high resolution 4K clinical displays.

To guarantee uninterrupted clinical operation, an autonomous hardware watchdog was engineered as a background failsafe loop. Upon detecting a USB or Bluetooth serial interruption, the watchdog autonomously reestablishes the COM port connection without requiring manual clinician intervention.

III. CLINICAL ASSESSMENT AND DATA PIPELINE

To accommodate both informal home use and rigorous clinical evaluation, the software's Finite State Machine (FSM) is organized into three distinct operational modes, as summarized in Table I.

TABLE I
FSM OPERATIONAL MODES SUMMARY

Mode	Primary Function	Output
Practice	Real-time biofeedback; unrecorded telemetry	Live digital twin
Record	Formal assessment; angular deviation logging	Clinical report card
Data Extraction	One click telemetry export	CSV / JSON dataset

A. Practice Mode (Real-Time Biofeedback)

Designed primarily for patient familiarization and home safety, Practice Mode provides continuous unrecorded telemetry. The patient's live digital twin is rendered in real time alongside an expert baseline *guide* avatar representing ideal execution of standardized clinical movements such as Tadasana or the Back-Scratch Test [10]. This mode leverages proprioceptive visual feedback, allowing users to self correct postural deviations and build motor memory without the pressure of clinical scoring.

B. Record Mode (Automated Diagnostics)

Utilized for formal clinical assessments, Record Mode activates the diagnostic engine. As the user follows the guide avatar, the system calculates and logs the precise 3D angular deviation $\Delta\theta$ between the live patient's digital twin and the expert guide avatar across all exercise phases. Upon completion, the FSM synthesizes this data into an automated clinical report card that actively flags potential mobility restrictions, localized angular discrepancies, and postural deformities [11], [12].

C. Data Extraction Mode (Telemetry Export)

To bridge raw physical therapy data with advanced computational analysis, this mode leverages native Windows file browser integration. Clinicians execute a single click export of high fidelity timestamped kinematic datasets captured during Record Mode. Data is exported in standardized CSV or JSON

formats, detailing both 4D quaternions and 3D Euler angles for all sensors, providing a structured pipeline for downstream predictive AI analysis.

IV. RESULTS AND DISCUSSION

A. Hardware Reliability and Telemetry Stability

Experimental validation demonstrated high hardware reliability. The ESP32 and TCA9548A I²C multiplexer architecture successfully maintained a stable telemetry stream at 115200 baud across three independent IMU channels without data collision. The Unity 6 digital twin interpreted incoming 4D quaternion data with negligible visual latency. The constrained biomechanical model successfully rejected simulated sensor drift and magnetic noise, preventing Gimbal Lock during complex movements including overhead shoulder flexion.

B. Software Deployment and Portability

The system was successfully deployed as a highly portable Windows application. The plug and play ZIP distribution was verified on secondary host machines with no installation dependencies. Adaptive resolution scaling maintained a proportionate interface across all tested hardware. Native OS integration successfully exported structured timestamped kinematic datasets from Data Extraction Mode, confirming the architecture's readiness for real world clinical data harvesting.

C. Live System Demonstration and Open Source Repository

The complete open source repository containing the source code, hardware schematics, and Unity application, alongside a live video demonstration showcasing the real time digital twin and clinical mode transitions, are available at:

GitHub Repository:

<https://github.com/RJariwala/SynchroKin>

Live Demonstration:

<https://www.youtube.com/watch?v=wV1gzhG-ApU>

V. CONCLUSION AND FUTURE WORK

This paper presents a low cost wearable IMU system bridging the critical gap between independent home safety and objective clinical diagnostics. By combining edge processed 4D quaternion mathematics with a heavily constrained Unity 6 digital twin, the system provides both preventative visual biofeedback for patients and high fidelity kinematic telemetry for practitioners.

Future work will focus on three directions. First, the native CSV/JSON export pipeline will be leveraged to aggregate anonymized kinematic data from diverse patient populations to train predictive machine learning models capable of identifying early onset musculoskeletal deformities before macroscopic clinical symptoms manifest. Second, rigorous IRB-approved comparative trials against industry standard optical motion capture systems (e.g., Vicon) will be conducted in hospital settings to statistically quantify angular accuracy (target: RMSE < 5°) and support formal medical device certification. Third, the hardware will evolve from rigid 3D

printed housings into flexible conductive smart garments to maximize patient compliance during extended unsupervised home monitoring.

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